# Coding Assignment: `yfinance` + Cleaning + Visualization + ML (30-day window)

## Objectives

1. Fetch 5 years of daily `Close` prices for one stock using `yfinance`.
2. Clean the data (remove all `NA` values).
3. Visualize the `Close` series.
4. **Supervised task (regression):** using a 30-day rolling window, train one supervised model of your choice to predict the next-day price.
5. **Unsupervised task (direction):** using a 30-day rolling window, train one unsupervised model of your choice to predict next-day movement (Up/Down).

## Setup

Install the required libraries:

```
pip install yfinance pandas scikit-learn matplotlib
```

## Step-by-Step Tasks

1. **Pick a ticker.** Choose any active stock (e.g., `AAPL`, `MSFT`, `TSLA`) and document your choice in a Markdown cell.
2. **Download and clean.** Fetch 5-year daily `Close` prices and remove all `NA` values.
3. **Visualize the series.**
   - Plot the entire daily `Close` price over the 5-year period.
   - Include: title (company name + ticker), labeled axes, grid lines, and a legend.
   - (Optional) Add 50-day and 200-day simple moving averages.
4. **Feature engineering (rolling 30 days).**
   - Construct features using the previous 30 trading days.
   - Recommended baseline feature sets (choose at least one):
     - **Lag closes:**
       $$[C_{t-29}, \ldots, C_t]$$
     - **Lag returns:**
       $$r_t = \frac{C_t}{C_{t-1}} - 1$$
     - (Optional) include technical indicators such as SMA(5/10/20), RSI, etc.

5. **Supervised regression task.**
   - Target: next-day close price $C_{t+1}$.
   - Use a time-aware split (e.g., first 80% train, last 20% test).
   - Train one supervised model (e.g., `Ridge`, `RandomForestRegressor`, `XGBRegressor`, etc.).
   - Report evaluation metrics: MAE, MAE, RMSE, and Runtime on the test set.
   - Use the most recent 30-day window to predict the next day's price.
   - **Visualization requirement:**
     - Plot *predicted vs. actual* next-day prices on the test set.
     - Include a diagonal reference line ($y = x$) to show prediction accuracy.
     - Title the figure clearly (e.g., "Predicted vs. Actual Close Prices — Supervised Model").

6. **Unsupervised direction prediction.**
   - Objective: predict the next-day direction (Up/Down) based on

     $$C_{t+1} - C_t.$$

   - Train one unsupervised model (e.g., `KMeans(n_clusters=2)`).
   - Recommended approach:
     a. Fit the model on 30-day feature windows $X_t$.
     b. For each cluster, compute the mean next-day return on the training set and assign:

        $$\text{cluster} \to \begin{cases} \text{Up}, & \text{if mean return} > 0, \\ \text{Down}, & \text{otherwise.} \end{cases}$$

     c. Predict the cluster for the latest 30-day window and map it to Up/Down.
     d. Evaluate performance on the test set using accuracy or F1-score.
   - **Visualization requirement:**
     - Plot the predicted vs. actual *movement direction* (Up/Down) on the test set.
     - Use different colors (e.g., green for Up, red for Down).
     - Clearly label the axes and title (e.g., "Predicted vs. Actual Next-Day Movement — Unsupervised Model").

# Visualization Summary (All Students Must Include)

Each submission must contain at least **three visualizations**:

(a) A 5-year time series plot of the raw `Close` price.

(b) A scatter or line plot of **predicted vs. actual next-day prices** (supervised model).

(c) A categorical plot comparing **predicted vs. actual movement directions** (unsupervised model).

Each figure must have:

- A clear and descriptive title.
- Labeled axes with units where applicable.
- Grid lines and a readable legend.